

Homework 6: Bayesian Inference

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1. LINK TO CODE

The code used in this homework can be found here:
[HW6_Bayesian.ipynb](#).

2. QUESTION 1

2.1. Part a

We are given a prior distribution of Beta(2, 5). After updating with 3 heads and 2 tails, the posterior becomes

$$\text{Beta}(2 + 3, 5 + 2) = \boxed{\text{Beta}(5, 7)}.$$

2.2. Part b

For 4 final flips, the number of ways to obtain exactly 2 heads is given by the binomial coefficient

$$\boxed{\binom{4}{2}}.$$

The corresponding binomial probability expression is

$$P(h | \theta) = \binom{n}{h} \theta^h (1 - \theta)^{n-h}.$$

Here, $n = 4$ and $h = 2$.

2.3. Part c

From part (a), the updated posterior is

$$\theta | D \sim \text{Beta}(5, 7),$$

so its probability density function is

$$p(\theta | D) = \frac{\Gamma(12)}{\Gamma(5)\Gamma(7)} \theta^4 (1 - \theta)^6.$$

Let X be the number of heads in the final 4 flips. Then

$$P(X = 2 | D) = \int_0^1 P(X = 2 | \theta) p(\theta | D) d\theta.$$

Using the binomial probability for exactly 2 heads in 4 flips,

$$P(X = 2 | \theta) = \binom{4}{2} \theta^2 (1 - \theta)^2,$$

we obtain

$$\begin{aligned} P(X = 2 | D) &= \int_0^1 \binom{4}{2} \theta^2 (1 - \theta)^2 \frac{\Gamma(12)}{\Gamma(5)\Gamma(7)} \theta^4 (1 - \theta)^6 d\theta \\ &= \binom{4}{2} \frac{\Gamma(12)}{\Gamma(5)\Gamma(7)} \int_0^1 \theta^6 (1 - \theta)^8 d\theta \\ &= \binom{4}{2} \frac{\Gamma(12)}{\Gamma(5)\Gamma(7)} B(7, 9) \\ &= \binom{4}{2} \frac{\Gamma(12)}{\Gamma(5)\Gamma(7)} \frac{\Gamma(7)\Gamma(9)}{\Gamma(16)} \\ &= 6 \cdot \frac{11! 8!}{4! 15!} \\ &= \frac{4}{13} \approx 0.3077. \end{aligned}$$

Therefore, the exact probability of getting 2 heads in the final 4 flips is

$$\boxed{\frac{4}{13} \approx 0.3077},$$

or about 30.77%.

3. QUESTION 2

3.1. Part a

The Bayesian inference code used in this homework is available at [HW6_Bayesian.ipynb](#).

3.2. Part b

Figure 1 shows the posterior probability distribution for the coin-bias parameter. The posterior obtained from the full dataset is centered away from $\theta = 0.5$, indicating that the coin is not balanced.

3.3. Part c

Figure 1 also shows the posteriors obtained using 5, 50, and 500 flips. As the number of flips increases, the posterior becomes narrower and more stable, indicating that the inferred probability is converging as more data are included.

3.4. Part d

Using different priors and different amounts of data produces distinct posteriors, as shown in Figure 2. The

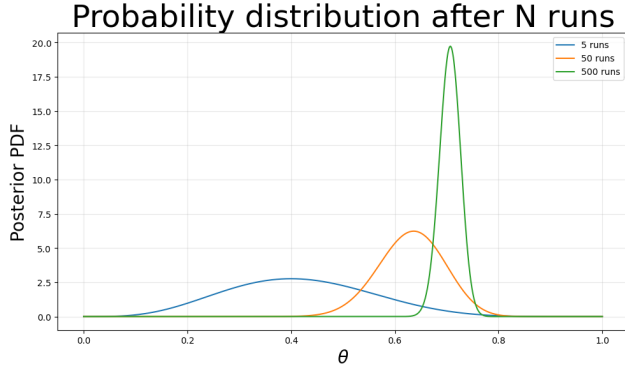


Figure 1. Posterior probability distributions for the coin-bias parameter.

prior has the strongest influence when only a small number of flips are used. As the number of flips increases, the data dominate and the posteriors from different priors become more similar.

4. QUESTION 3

4.1. Part a

The randomly generated dataset is shown in Figure 3. Points were generated uniformly in the unit square, and points satisfying $x^2 + y^2 \leq 1$ fall inside the quarter circle.

4.2. Part b

Figure 4 shows the posterior probability distributions for π using 5, 50, 500, and 1000 datapoints. As the number of datapoints increases, the estimate of π improves. In my run, the best estimate is $\hat{\pi} = 3.1377$, corresponding to a relative error of 0.12%.

4.3. Part c

Using priors similar to those in Question 2 produces similar trends, as shown in Figure 5. The prior has a proportionally larger influence when the number of datapoints is small. As more datapoints are included, the posterior becomes less sensitive to the prior and converges toward the true value of π .

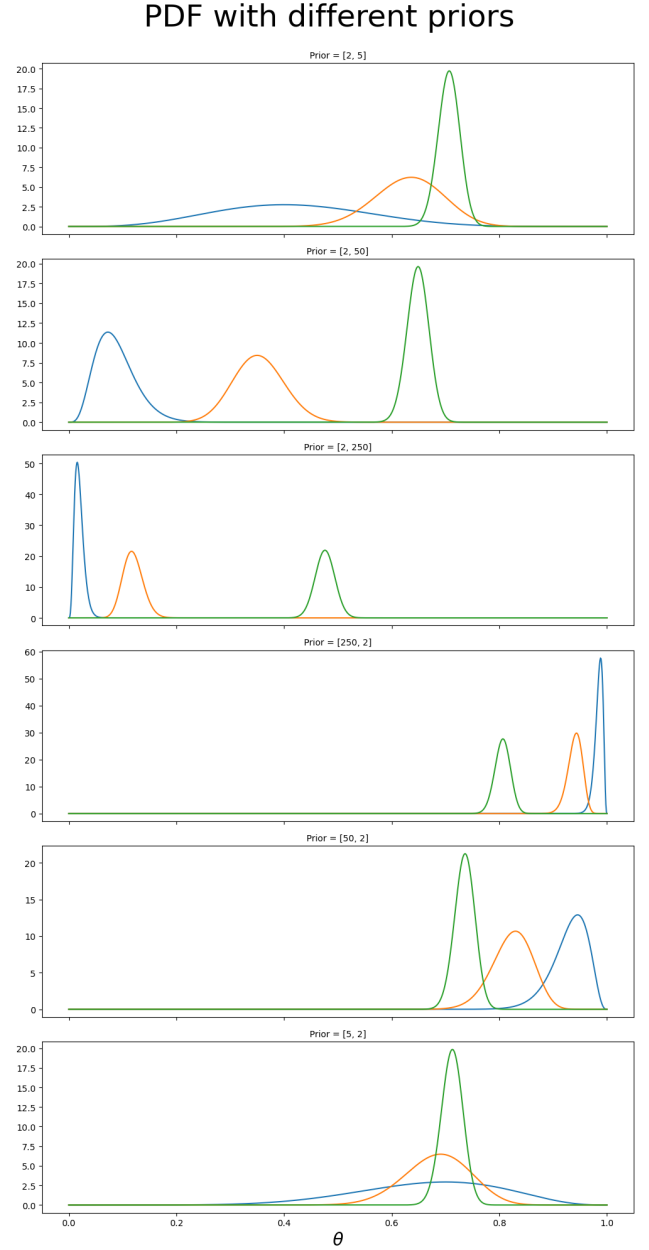


Figure 2. Posterior probability distributions for different prior choices and different numbers of coin flips.

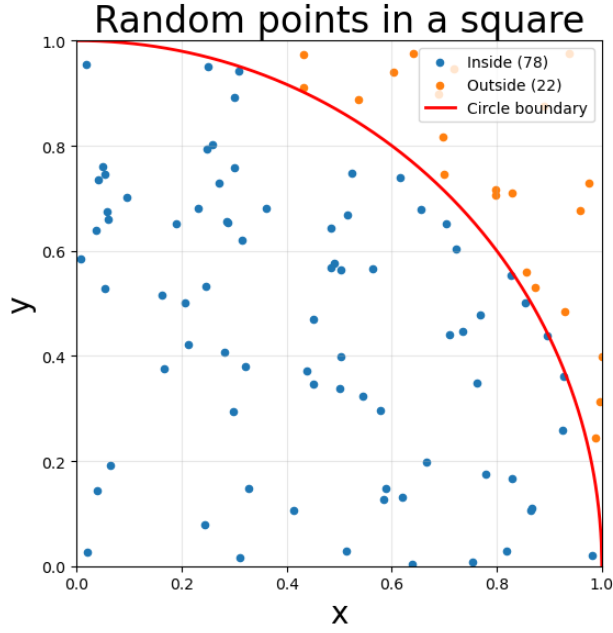


Figure 3. Randomly generated points in the unit square and the quarter-circle boundary.

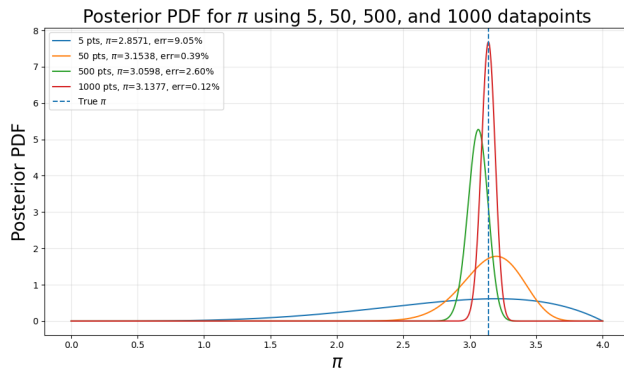


Figure 4. Posterior probability distributions for π using different numbers of datapoints.

PDF with different priors

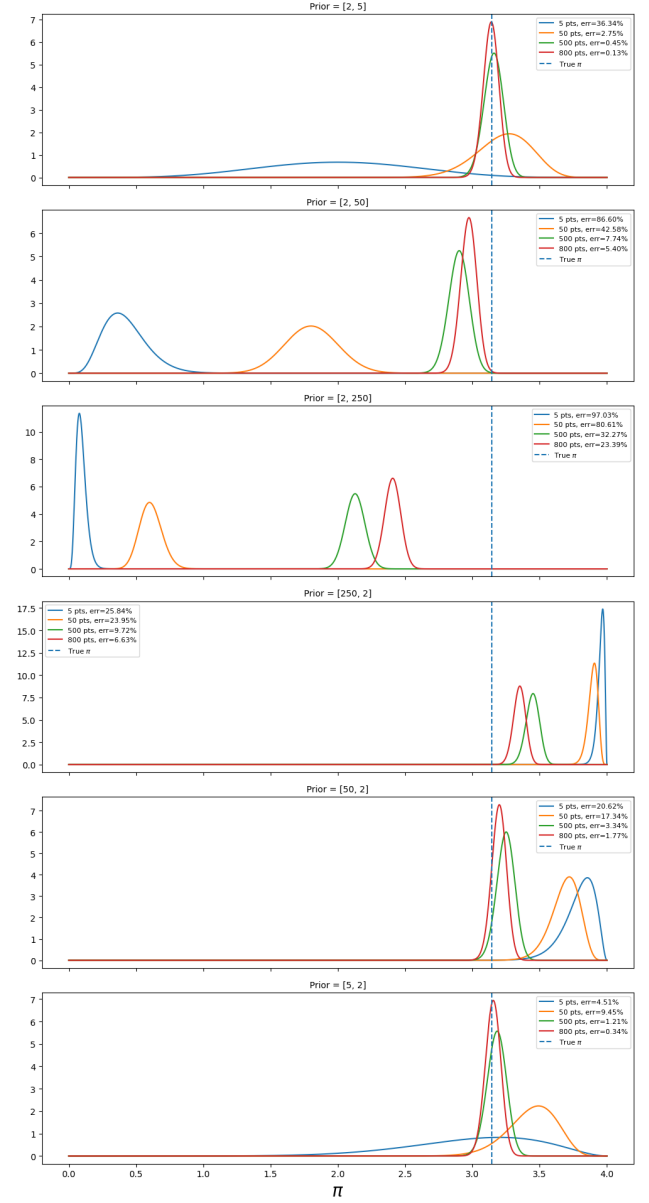


Figure 5. Posterior probability distributions for π using different priors.